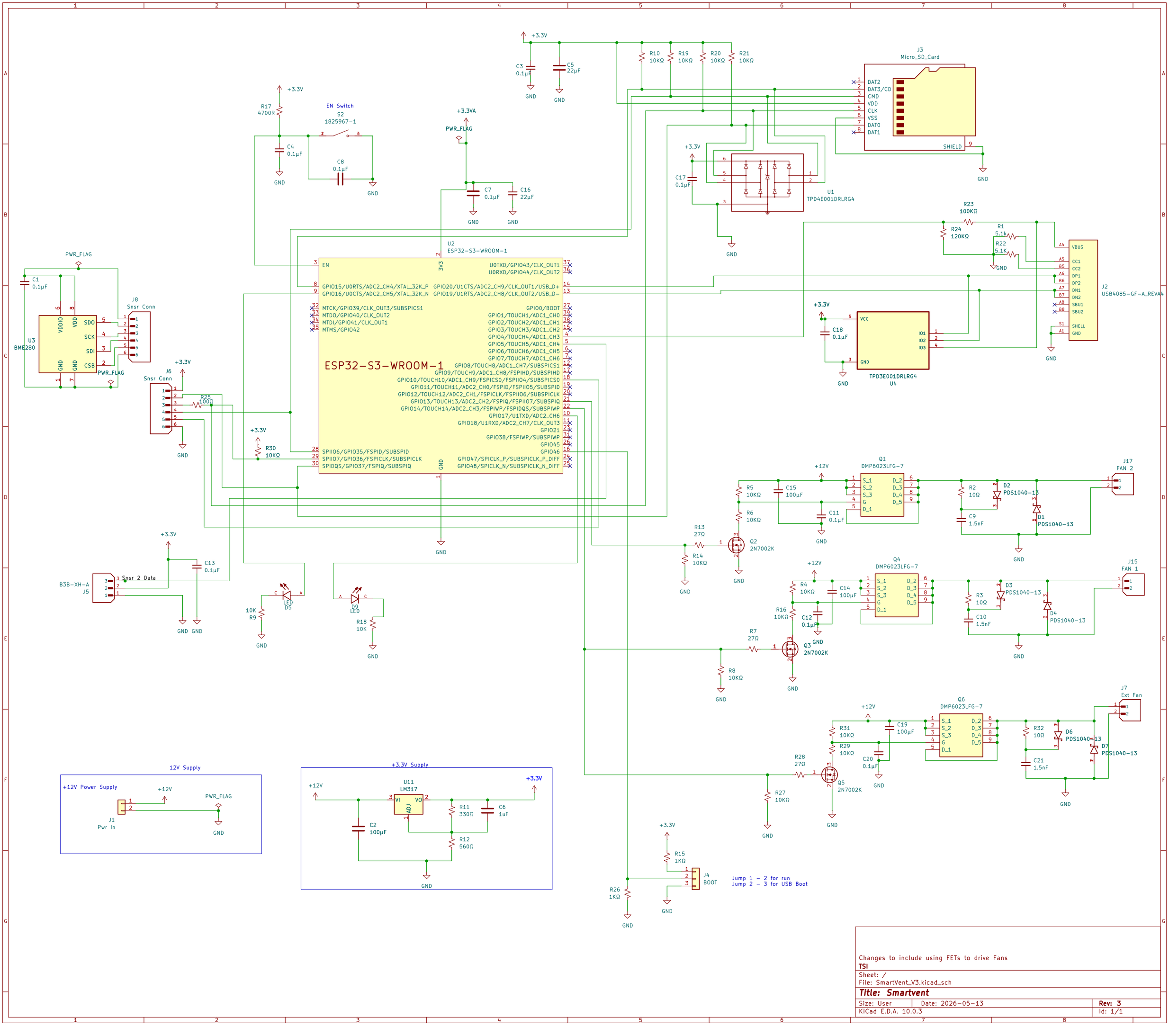

ESP32-S3-WROOM-1
U2
ESP32-S3-WROOM-1
EN
3V3
GND
GPIO15/U0RTS/ADC2_CH4/XTAL_32K_P
GPIO16/U0CTS/ADC2_CH5/XTAL_32K_N
GPIO20/U1CTS/ADC2_CH9/CLK_OUT1/USB_D+
GPIO19/U1RTS/ADC2_CH8/CLK_OUT2/USB_D-
U0TXD/GPIO43/CLK_OUT1
U0RXD/GPIO44/CLK_OUT2
MTCK/GPIO39/CLK_OUT3/SUBSPICS1
MTDO/GPIO40/CLK_OUT2
MTDI/GPIO41/CLK_OUT1
MTMS/GPIO42
GPIO0/BOOT
GPIO1/TOUCH1/ADC1_CH0
GPIO2/TOUCH2/ADC1_CH1
GPIO3/TOUCH3/ADC1_CH2
GPIO4/TOUCH4/ADC1_CH3
GPIO5/TOUCH5/ADC1_CH4
GPIO6/TOUCH6/ADC1_CH5
GPIO7/TOUCH7/ADC1_CH6
GPIO8/TOUCH8/ADC1_CH7/SUBSPICS1
GPIO9/TOUCH9/ADC1_CH8/FSPIHD/SUBSPIHD
GPIO10/TOUCH10/ADC1_CH9/FSPICS0/FSPIIO4/SUBSPICS0
GPIO11/TOUCH11/ADC2_CH0/FSPID/FSPIIO5/SUBSPID
GPIO12/TOUCH12/ADC2_CH1/FSPICLK/FSPIIO6/SUBSPICLK
GPIO13/TOUCH13/ADC2_CH2/FSPIQ/FSPIIO7/SUBSPIQ
GPIO14/TOUCH14/ADC2_CH3/FSPIWP/FSPIDQS/SUBSPIWP
GPIO17/U1TXD/ADC2_CH6
GPIO18/U1RXD/ADC2_CH7/CLK_OUT3
GPIO21
GPIO38/FSPIWP/SUBSPIWP
GPIO45
GPIO46
GPIO47/SPICLK_P/SUBSPICLK_P_DIFF
GPIO48/SPICLK_N/SUBSPICLK_N_DIFF
SPIIO6/GPIO35/FSPID/SUBSPID
SPIIO7/GPIO36/FSPICLK/SUBSPICLK
SPIDQS/GPIO37/FSPIQ/SUBSPIQ
+3.3V
C3 0.1µF
C5 22µF
R10 10KΩ
R19 10KΩ
R20 10KΩ
R21 10KΩ
J3
Micro_SD_Card
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
R17 4700R
EN Switch
S2
1825967-1
C4 0.1µF
C8 0.1µF
+3.3VA
PWR_FLAG
C7 0.1µF
C16 22µF
C17 0.1µF
U1
TPD4E001DRLRG4
R23 100KΩ
R24 120KΩ
R1 5.1k
R22 5.1K
J2
USB4085-GF-A_REVA4
VBUS
CC1
CC2
DP1
DP2
DN1
DN2
SBU1
SBU2
SHELL
GND
C18 0.1µF
U4
TPD3E001DRLRG4
VCC
IO1
IO2
IO3
PWR_FLAG
C1 0.1µF
J8
Snsr Conn
U3
BME280
VDDIO
VDD
SDO
SCK
SDI
CSB
GND
J6
Snsr Conn
R25 100Ω
R30 10KΩ
Q1
DMP6023LFG-7
R5 10KΩ
C15 100µF
R6 10KΩ
C11 0.1µF
R2 10Ω
C9 1.5nF
D2
PDS1040-13
D1
PDS1040-13
J17
FAN 2
R13 27Ω
Q2
2N7002K
R14 10KΩ
Q4
DMP6023LFG-7
R4 10KΩ
C14 100µF
R16 10KΩ
C12 0.1µF
R3 10Ω
C10 1.5nF
D3
PDS1040-13
D4
PDS1040-13
J15
FAN 1
R7 27Ω
Q3
2N7002K
R8 10KΩ
J7
Ext Fan
Q6
DMP6023LFG-7
R31 10KΩ
R29 10KΩ
C19 100µF
C20 0.1µF
R32 10Ω
C21 1.5nF
D6
PDS1040-13
D7
PDS1040-13
R28 27Ω
Q5
2N7002K
R27 10KΩ
B3B-XH-A
J5
Snsr 2 Data
C13 0.1µF
10K
R9
LED
D5
D9
LED
R18 10K
12V Supply
+12V Power Supply
J1
Pwr In
+3.3V Supply
U11
LM317
VI
VO
ADJ
R11 330Ω
C6 1uF
C2 100µF
R12 560Ω
R15 1KΩ
J4
BOOT
Jump 1 - 2 for run
Jump 2 - 3 for USB Boot
R26 1KΩ
Changes to include using FETs to drive Fans
TSI
Sheet: /
File: SmartVent_V3.kicad_sch
Title: Smartvent
Size: User
Date: 2026-05-13
Rev: 3
KiCad E.D.A. 10.0.3
Id: 1/1